

E-04-LEMMA-C-quant v1: quantitative audit of C_S

ZFL Lab

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Goal: Determine C_{quant} in $|W| \leq C_{\text{quant}} \|u\|_3 \|\nabla \omega\|_2^2$.

1. Derivation exact:

$$W := \int_{R^3} \omega^T S \omega \, dx = \int \omega_i \partial_j u_i \omega_j \, dx.$$

$$\text{IBP (div-free): } W = - \int u_i (\partial_j \omega_i) \omega_j \, dx.$$

$$\text{Holder: } |W| \leq \|u\|_{L^3} \|\nabla \omega\|_{L^2} \|\omega\|_{L^6}.$$

$$\text{Sobolev } R^3: \|\omega\|_{L^6} \leq C_S \|\nabla \omega\|_{L^2}.$$

Hence:

$$\boxed{|W| \leq C_S \|u\|_3 \|\nabla \omega\|_2^2.}$$

Thus $C_{\text{quant}} = C_S$.

2. Conventions: Domain R^3 , viscosity $\nu > 0$ explicit, norms L^3, L^2, L^6 standard.

3. Value of C_S :

Sobolev sharp constant (Talenti): $\|f\|_{L^6} \leq C_S^{\text{sharp}} \|\nabla f\|_{L^2}$, $C_S^{\text{sharp}} = \pi^{-1/2} \cdot (\text{explicit})$; classical bound $C_S \leq \frac{2^{2/3}}{\sqrt{3}\pi^{2/3}}$ etc depending normalization. We distinguish: $C_S^{\text{available}} = \text{explicit non-sharp constant used}$, $C_S^{\text{sharp}} = \text{optimal constant}$.

For this lemma we keep C_S symbolic, audit says: $C_{\text{quant}} = C_S$ PROVEN, numeric value depends on Sobolev convention.

4. Consequence:

$$\text{If } M := \sup_{t < T_*} \|u(t)\|_3, |W| \leq C_S M \|\nabla \omega\|_2^2, \text{ enstrophy: } \frac{1}{2} \frac{d}{dt} \|\omega\|_2^2 \leq (C_S M - \nu) \|\nabla \omega\|_2^2.$$

$$\boxed{M < \nu/C_S \implies T_* = \infty.}$$

Critical threshold: $M_{\text{crit}} := \nu/C_S^{\text{sharp}}$ if sharp constant known.

Classification:

Calculation of C_S as source of C_{quant} PROVEN

Threshold $M < \nu/C_S \implies T_* = \infty$ PROVEN

Existence/sharpness near threshold OPEN

No new C_F . Global state: $\mathcal{C}_{\text{PROVEN}} = \{C_E^{\text{weak}}, C_{L^3}\}$ plus $|W| \leq C_S \|u\|_3 \|\nabla \omega\|_2^2$ RELATION PROVEN $CM < \nu$ THRESHOLD PROVEN C_S quantitative audit done.

Not declaring M_{crit} reachable nor sharp.